Assignment 4

Exercise 1

a. Flavour = { Strawberry, Anchovy }
Shape = { Round, Square }
Wrapper = { Red, Brown }

- $P(S) = 0.7$, $P(A) = 0.3$
- $P(Round|S) = 0.8$ $P(Square|A) = 0.9$
  $P(Square|S) = 0.2$ $P(Round|A) = 0.1$
- $P(Red|S) = 0.8$ $P(Red|A) = 0.9$
  $P(Brown|S) = 0.2$ $P(Brown|A) = 0.1$

Looks like shape and colour wrapper is dependant on flavour, but independent of each other.
So. network (iii)

(i) States that wrapper & shape affect flavour which is not true as flavour influences the other variables. So this network cannot represent the dependencies.

(ii) This states wrapper affects shape, which makes sense as both are influenced by flavour. It can represent dependencies

(iii) As stated earlier (iii) represents the dependencies best.

b. (iii) is best as it matches the real causal structure and does not have a cyclic dependency like (ii). It makes deducing the numbers required by the conditional probability tables easier.

c. In (i) There is no arrow from and to wrapper to shape So they are Independent of each other.

d) $P(Red) = P(Red|S) \cdot P(S) + P(Red|A) \cdot P(A)$
$= 0.8 \cdot 0.7 + 0.1 \cdot 0.3$
$= \underline{0.59}$

e) $P(S|Round, Red) = \frac{P(Round, Red \cap S)}{P(Round, Red)} = \frac{P(Round|S) \cdot P(Red|S) \cdot P(S)}{P(Round|S) \cdot P(Red|S) \cdot P(S) + P(Round|A) \cdot P(Red|A) \cdot P(A)}$

$= \frac{0.8 \cdot 0.8 \cdot 0.7}{0.8 \cdot 0.8 \cdot 0.7 + 0.1 \cdot 0.1 \cdot 0.3} = \frac{0.488}{0.488 + 0.003} = \underline{0,99335}$

f. $E(V) = P(S) \cdot s + P(A) \cdot a$
$= \underline{0.7s + 0.3a}$

Exercise 2

$U(x) = -e^{-x/R}$

a) 2 options; a) Win 500 dollars, b) 60% chance of winning \$5000 or 40% of winning nothing.

$U(x)$ for a) $= U(500) = -e^{-500/500} = -e^{-1} = -\frac{1}{e} = \underline{-0,36}$

b) $0.6 \cdot -e^{-5000/500} + 0.4 \cdot -e^{-1} = \underline{-0.40}$

$U(500)$ is larger so if Mary acted rationally she would choose option a.

b) ⓐ $U(100) = -e^{-100/R}$

ⓑ $0,50 \cdot U(500) + 0,50 \cdot U(0) = -.50 \cdot e^{-500/R} - .50 \cdot e^{0} = -0,5 \cdot e^{-500/R} - 0,5$

We can equate them to each other as "indifference" means they are equal in utility. We can then find risk.

$-e^{-100/R} = -0,5 \cdot e^{-500/R} - 0,5$

$0,5 \cdot \frac{1}{e^{500/R}} - \frac{1}{e^{100/R}} + 0,5 = 0$

$e^{-100/R} = 0{,}5e^{-500/R} + 0{,}5$

If $x = e^{-100/R}$, then we can solve equation with substitution:

$$x = 0{,}5 \cdot x^5 + 0{,}5$$

$$\Rightarrow 2x = x^5 + 1$$

$$x - x^5 - 2x + 1 = 0$$

$x = 1$ is a solution

$$1 = e^{-100/R}$$

$$\ln(1) = \ln(e^{-100/R})$$

$$\frac{-100}{R} = 0 \Rightarrow X$$

This is not valid

$$x = 0{,}518$$

$$e^{-100/R} = 0{,}518$$

$$\ln(e^{-100/R}) = \ln(0{,}518)$$

$$\frac{-100}{R} = \ln(0{,}518)$$

$$R = \frac{-100}{\ln(0{,}518)} = 152{,}026$$